RESPONSIBLE DATA CARD

Customer Churn Prediction Using Machine Learning

1. Dataset Overview

Dataset Name: Customer Churn Training Data
Project: Customer Churn Prediction
Purpose: To analyze customer information and predict whether a customer may stop using a company's services.

The dataset is intended for developing and evaluating a Machine Learning model for customer churn prediction.

2. Intended Use

The dataset will be used to:

- Understand patterns related to customer churn.
- Train a baseline classification model.
- Evaluate the model's ability to identify potential churn.
- Explore ways businesses may improve customer retention.

The model is intended to support human decision-making, not replace it.

3. Data Types

The dataset may contain customer information such as:

- Customer identifiers.
- Customer account or service details.
- Usage or billing information.
- Customer tenure.
- Churn status (the target variable).

Note: The actual columns and data types must be confirmed by inspecting the supplied dataset.

4. Data Quality

Before training the model, the dataset should be checked for:

- Missing values.
- Duplicate records.
- Incorrect or inconsistent values.
- Imbalanced churn and non-churn classes.
- Unnecessary or sensitive information.

Any data-cleaning steps should be documented.

5. Privacy and Security

- Use only the data needed for the project.
- Do not publish personal or identifiable customer information.
- Store the dataset securely.
- Remove or mask personal identifiers where appropriate.
- Share only permitted files in the public GitHub repository.

6. Fairness Considerations

The model may perform differently for different customer groups. Its performance should be evaluated carefully, where suitable group information is available and its use is permitted.

Sensitive attributes should not be used without a valid justification and appropriate authorization.

7. Limitations

- The dataset may not represent all customers.
- Historical patterns may not predict future behaviour accurately.
- Missing or biased data can affect model performance.
- Predictions may contain false positives and false negatives.
- Customer behaviour may change over time.

8. Responsible Use

The model should be evaluated before real-world use. Predictions should be reviewed by a human, and customers should not be unfairly penalized based only on the model's output.

9. Conclusion

Responsible data handling is essential for building a reliable customer churn prediction system. Data quality, privacy, fairness, transparency, and human oversight should be considered throughout the project.